606027 ← Test Booklet No.

ENGLISH

Test Booklet Code

12

NEET RANKERS TEST SERIES-2025

Do not open this Test Booklet until you are asked to do so.

Important Instructions:

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours 20 minutes** duration and the Test Booklet contains **200** multiple-choice questions (four options with a single correct answer) from **Physics, Chemistry and Biology (Botany and Zoology)**. **50** questions in each subject are divided into **two Sections (A and B)** as per details given below.
   - (a) **Section A** shall consist of **35 (Thirty-five)** Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 and 185). All questions are compulsory.
   - (b) **Section B** shall consist of **15 (Fifteen)** questions in **each subject (Question** Nos - 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to **attempt any 10 (Ten)** questions out of **15 (Fifteen)** in each **subject**.

   Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, **the first ten questions answered by the candidate shall be evaluated**
3. Each question carries 4 marks. For each correct response, the candidate will get **4 marks**. For each incorrect response, **one** mark will be deducted from the total scores. **The maximum marks are 720.**
4. **Use Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
5. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
6. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
7. **The CODE for this Booklet is 12. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet.** In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
8. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
9. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
10. Each candidate must show on-demand his/her Admit Card to the Invigilator
11. No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
12. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet twice. **Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.**
13. Use of Electronic/Manual Calculator is prohibited.
14. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
15. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
16. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
17. Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): ____________________

Roll Number: In figures ____________________

: In words ____________________

Centre of Examination (in Capitals): ____________________

Candidate's Signature: ____________________ **Invigilator's Signature:** ____________________

Facsimile Signature stamp of Centre Superintendent

Physics : Section-A (Q. No. 1 to 35)

1. The ratio of amplitude of magnetic field to the amplitude of electric field for an electromagnetic wave propagating in vacuum is equal to:

(1) the speed of light in vacuum

(2) reciprocal of speed of light in vacuum

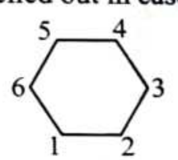
(3) the ratio of magnetic permeability to the electric susceptibility of vacuum

(4) unity

2. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): Four Point masses each of mass m are placed at points 1, 2, 3 and 6 of a regular hexagon of side a. Then the gravitational field at the centre of hexagon is $\frac{Gm}{a^2}$.

Reason (R): The field strength due to masses at 3 and 6 are cancelled out in case of regular hexagon.



In the light of the above statements, choose the **correct** answer from the options given below:

(1) **A** is true but **R** is false.

(2) **A** is false but **R** is true.

(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.

(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

3. A particle moves along the X-axis from $x = 0$ to $x = 5m$ under the influence of a force given by $F = 10 - 2x + 3x^2$. Work done in the process is:

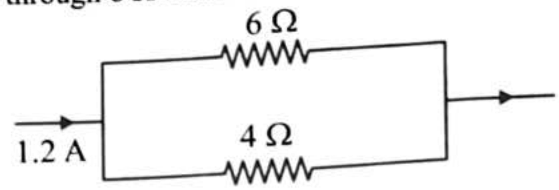
(1) 70 units

(2) 270 units

(3) 35 units

(4) 150 units

4. In the figure given below, the current passing through 6 Ω resistor is:



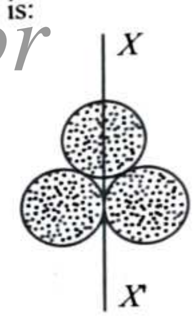
(1) 0.40 A

(2) 0.48 A

(3) 0.72 A

(4) 0.80 A

5. Three identical spherical shells, each of mass m and radius r are placed as shown in figure. Consider an axis XX', which is touching to two shells and passing through diameter of third shell. Moment of inertia of the system consisting of these three spherical shells about XX' axis is:



(1) $\frac{11}{5}mr^2$

(2) $3\ mr^2$

(3) $\frac{16}{5}mr^2$

(4) $4\ mr^2$

6. The total energy of an electron in an atom in an orbit is –3.4 eV. Its kinetic and potential energies are, respectively

(1) –3.4 eV, –6.8 eV

(2) 3.4 eV, –6.8 eV

(3) 3.4 eV, 3.4 eV

(4) –3.4 eV, –3.4 eV

7. The power of a biconvex lens is 10 D and the radius of curvature of each surface is 10 cm. Then, the refractive index of the material of the lens is:

(1) $\frac{4}{3}$

(2) $\frac{9}{8}$

(3) $\frac{5}{3}$

(4) $\frac{3}{2}$

8. In a transformer, the number of turns in primary and secondary are 500 and 2000, respectively. If current in primary is 48 A, the current in the secondary is:

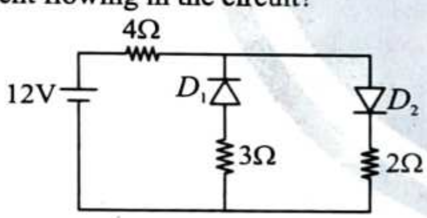
(1) 12 A

(2) 24 A

(3) 48 A

(4) 144 A

9. The circuit has two oppositely connected ideal diodes in parallel as shown in figure. What is the current flowing in the circuit?



(1) 1.33 A

(2) 1.71 A

(3) 2.00 A

(4) 2.31 A

10. The range of a projectile when launched at angle θ is same as when launched at angle 2θ. What is the value of θ?

(1) 15° (2) 30°

(3) 45° (4) 60°

11. The ratio of wavelength of the last line of Balmer series and the last line of Lyman series is:

(1) 2

(2) 1

(3) 4

(4) 0.5

12. What is the phase difference, at a given instant of time, between two particles 25 m apart, when the wave $y(x, t) = 0.03 \sin \pi (2t - 0.01x)$ travels in a medium?

(1) $\frac{\pi}{8}$

(2) $\frac{\pi}{4}$

(3) $\frac{\pi}{2}$

(4) π

13. Given below are two statements:

Two identical balls are given charge q each. They are suspended from a common point by two insulating threads of length l each.

Statement-I: In equilibrium, the maximum angle between the threads is 180°. (Ignore gravity).

Statement-II: In equilibrium, tension in the threads is $T = \frac{1}{4\pi\varepsilon_0}\frac{q \cdot q}{l^2}$

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement-I is correct but Statement-II is incorrect.

(2) Statement-I is incorrect but Statement-II is correct.

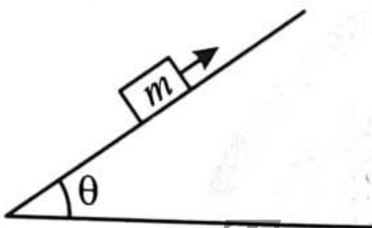
(3) Both Statement-I and Statement-II are correct.

(4) Both Statement-I and Statement-II are incorrect.

14. The magnetic flux ϕ (in weber) in a closed circuit of resistance 10 Ω varies with time t (in second) according to equation $\phi = 6t^2 - 5t + 1$. The magnitude of induced current at $t = 0.25$ s is:

(1) 1.2 A
(2) 0.8 A
(3) 0.6 A
(4) 0.2 A

15. A block of mass m is thrown upwards with some initial velocity as shown in figure. Match **List-I** with **List-II** for the block (All the symbols have usual meaning)



	List-I		List-II
(A)	Net force in horizontal direction	(I)	Zero
(B)	Net force in vertical direction	(II)	$m(g\sin\theta + \mu g\cos\theta)$
(C)	Net force along the plane	(III)	$m(g\sin\theta.\cos\theta + \mu g\cos^2\theta)$
(D)	Net force perpendicular to plane	(IV)	$m(g\sin^2\theta + \mu g\sin\theta\cos\theta)$

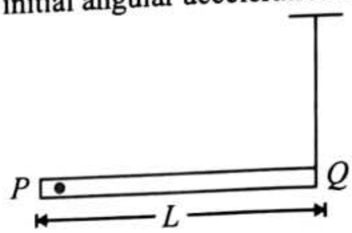
Choose the **correct** answer from the options given below:

(1) A-IV, B-I, C-II, D-III
(2) A-I, B-IV, C-III, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-III, C-I, D-II

16. Monochromatic green light of wavelength 5×10^{-7} m illuminates a pair of slits 1 mm apart. The separation of bright lines on the interference pattern formed on a screen 2 m away is:

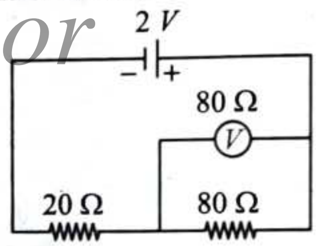
(1) 0.25 mm
(2) 0.1 mm
(3) 1.0 mm
(4) 0.01 mm

17. A rod PQ of mass M and length L is hinged at end P. The rod is kept horizontal by a massless string tied to point Q as shown in figure. When string is cut, the initial angular acceleration of the rod is:



(1) $\frac{3g}{2L}$ (2) $\frac{g}{L}$
(3) $\frac{2g}{L}$ (4) $\frac{2g}{3L}$

18. In the following circuit, the emf of the cell is 2 V and the internal resistance is negligible. The resistance of the voltmeter is 80 Ω. The reading of the voltmeter will be:



(1) 0.80 V
(2) 1.60 V
(3) 1.33 V
(4) 2.00 V

19. Stationary waves are formed when:

(1) two waves of equal amplitude and equal frequency travel along the same path in opposite directions.
(2) two waves of equal wavelength and equal amplitude travel along the same path with equal speeds in same directions.
(3) two waves of equal wavelength and equal phase travel along the same path with equal speed.
(4) two waves of equal amplitude and equal speed travel along the same path in opposite directions.

20. The de-Broglie wavelength of neutron at 27°C is λ. What will be its wavelength at 927°C.

(1) $\frac{\lambda}{2}$ (2) $\frac{\lambda}{3}$

(3) $\frac{\lambda}{4}$ (4) $\frac{\lambda}{9}$

21. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): Force cannot be added to pressure.

Reason (R): The dimensions of force and pressure are different.

In the light of the above statements, choose the **correct** answer from the options given below:

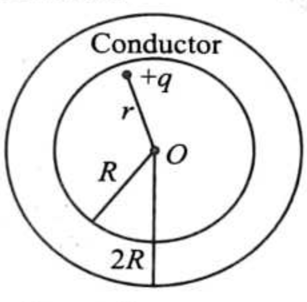
(1) **A** is true but **R** is false.

(2) **A** is false but **R** is true.

(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.

(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

22. A point charge q is place at a distance r from the centre O of an uncharged conducting spherical shell of inner radius R and outer radius $2R$. The distance $r < R$. The electric potential at the centre of the shell will be:



(1) $\frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r}-\frac{1}{2R}\right)$ (2) $\frac{q}{4\pi\varepsilon_0 r}$

(3) $\frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r}+\frac{1}{2R}\right)$ (4) $\frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r}-\frac{1}{R}\right)$

23. A coin placed on a rotating turn-table slips, when it is placed at a distance of 9 cm from the centre. If the angular velocity of the turn-table is tripled, it will just slip, if its distance from the centre is:

(1) 27 cm

(2) 9 cm

(3) 3 cm

(4) 1 cm

24. Two bodies of masses m_1 and m_2 have same momentum. The ratio of their KE is:

(1) $\sqrt{\frac{m_2}{m_1}}$

(2) $\sqrt{\frac{m_1}{m_2}}$

(3) $\frac{m_1}{m_2}$

(4) $\frac{m_2}{m_1}$

25. A proton and a deuteron, both having the same kinetic energy, enter perpendicularly into a uniform magnetic field B. For motion of proton and deuteron in circular path of radius R_p and R_d, respectively, the correct relation is:

(1) $R_d = \sqrt{2}R_p$

(2) $R_d = \frac{R_p}{\sqrt{2}}$

(3) $R_d = R_p$

(4) $R_d = 2R_p$

26. The wavelength of light in two liquids x and y is 3500 Å and 7000 Å respectively, then the critical angle of x relative to y will be:

(1) 60°

(2) 45°

(3) 30°

(4) 15°

27. Two parallel infinite line charges with linear charge densities $+\lambda$ C/m and $-\lambda$ C/m are placed at a distance of $2R$ in free space. What is the electric field mid-way between the two line charges?

(1) $\frac{2\lambda}{\pi\varepsilon_0 R}$ N/C (2) $\frac{\lambda}{\pi\varepsilon_0 R}$ N/C

(3) $\frac{\lambda}{2\pi\varepsilon_0 R}$ N/C (4) Zero

28. A gate has the following truth table

P	Q	R
1	1	1
1	0	0
0	1	0
0	0	0

The gate is:

(1) NOR (2) OR

(3) NAND (4) AND

29. Young's modulus of steel is 1.9×10^{11} Nm^{-2}. When expressed in CGS units of dyne/cm^2, it will be equal to ($1\ N = 10^5$ dyne, $1\ m^2 = 10^4\ cm^2$)

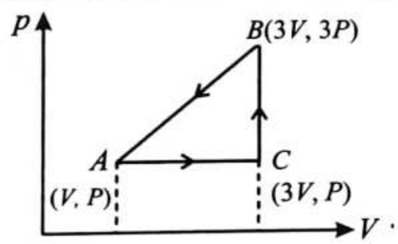
(1) 1.9×10^{10}

(2) 1.9×10^{11}

(3) 1.9×10^{12}

(4) 1.9×10^{13}

30. An ideal gas is taken around $ABCA$ as shown in the p-V diagram. The work done during the cycle is:



(1) $2pV$ (2) pV

(3) $\frac{pV}{2}$ (4) zero

31. When a charged particle enters a uniform magnetic field, then its kinetic energy.

(1) remains constant

(2) increases

(3) decreases

(4) becomes zero

32. For a gas $\gamma = \frac{5}{3}$ and 640 cc of this gas is suddenly compressed to 80 cc. If the initial pressure is p, then the final pressure will be

(1) $8p$

(2) $32p$

(3) $16p$

(4) $64p$

33. The interference patterns is obtained with two coherent light sources of intensity ratio n. In the interference pattern, the ratio $\frac{I_{max}-I_{min}}{I_{max}+I_{min}}$ will be:

(1) $\frac{\sqrt{n}}{n+1}$ (2) $\frac{2\sqrt{n}}{n+1}$

(3) $\frac{\sqrt{n}}{(n+1)^2}$ (4) $\frac{2\sqrt{n}}{(n+1)^2}$

34. Two charges of equal magnitudes and at a distance r exert a force F on each other. If the charges are halved and distance between them is doubled, then the new force acting on each charge is:

(1) $F/8$ (2) $F/4$

(3) $4F$ (4) $F/16$

35. A charge Q is enclosed by a Gaussian spherical surface of radius R. If the radius is doubled, then the outward electric flux will:

(1) be reduced to half

(2) remain the same

(3) be doubled

(4) increase four times

Physics : Section-B (Q. No. 36 to 50)

36. Which of the following statements are **true**?

A. A particle moving on a straight line may have increasing speed when acceleration is decreasing.

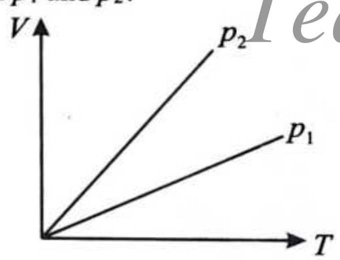
B. A body can have constant velocity with varying speed.

C. A body can have velocity without acceleration.

D. A body can have acceleration without having velocity.

(1) A and B
(2) A, B and C
(3) A, C and D
(4) B, C and D

37. In the adjacent V-T diagram what is the relation between p_1 and p_2?



(1) $p_2 = p_1$
(2) $p_2 > p_1$
(3) $p_2 < p_1$
(4) Cannot be predicted

38. The electric field associated with an electromagnetic wave in vacuum is given by $E = 40\cos(kz - 6 \times 10^8 t)\hat{i}$, where E, z and t are in volt m^{-1}, metre and second, respectively. The value of wave vector k is:

(1) 2 m^{-1}
(2) 0.5 m^{-1}
(3) 6 m^{-1}
(4) 3 m^{-1}

39. A simple harmonic oscillator consists of a particle of mass m and an ideal spring with spring constant k. The particle oscillates with a time period T. The spring is cut into two equal parts. If one part oscillates with the same particle, the time period will be:

(1) $2T$
(2) $\sqrt{2}T$
(3) $\frac{T}{\sqrt{2}}$
(4) $\frac{T}{2}$

40. Centre of mass of three particles of masses 1 kg, 2 kg, and 3 kg lies at the point (1, 2, 3) and centre of mass of another system of particles of total mass 3 kg lies at the point (–1, 3, –2). Where should we put a particle of mass 5 kg so that the centre of mass of the entire system lies at the centre of mass of Ist system?

(1) (0, 0, 0)
(2) (1, 3, 2)
(3) (–1, 2, 3)
(4) $\left(\frac{11}{5}, \frac{7}{5}, 6\right)$

41. In order to increases the sensitivity of a moving coil galvanometer one should decrease:

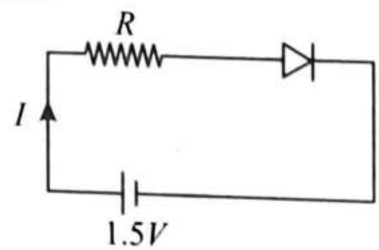
(1) the strength of its magnet
(2) the torsional constant of its suspension
(3) the number of turns in its coil
(4) the area of its coil

42. A sinusoidal voltage of peak value 300 V and an angular frequency ω = 400 rads^{-1} is applied to series L-C-R circuit, in which $R = 3\Omega$, L = 20 mH and C = 625μF. The peak value of current in the circuit is:

(1) $30\sqrt{2}\,A$
(2) $60\,A$
(3) $100\,A$
(4) $60\sqrt{2}\,A$

[Contd ..

43. The diode used at a constant potential drop of 0.5 V at all currents and maximum power rating of 100 mW. What resistance must be connected in series to diode, so that current in circuit is maximum?

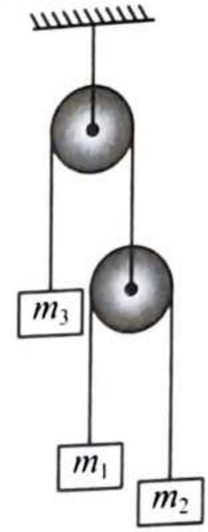


(1) 200 Ω
(2) 6.67 Ω
(3) 5 Ω
(4) 15 Ω

44. A force F is applied on a square plate of side L. If the percentage error in the determination of L is 2% and that in F is 4%. What is the permissible error in pressure?
(1) 8%
(2) 6%
(3) 4%
(4) 2%

45. A metallic rod of length l and cross-sectional area A is made of a material of Young modulus Y. if the rod is elongated by an amount y, then the work done is proportional to
(1) y
(2) $\frac{1}{y}$
(3) y^2
(4) $\frac{1}{y^2}$

46. In the figure, pulleys are smooth and strings are massless, $m_1 = 1$ kg and $m_2 = \frac{1}{3}$kg. To keep m_3 at rest, mass m_3 should be



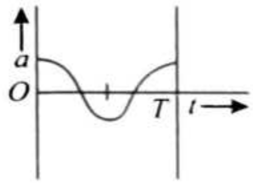
(1) 1 kg
(2) $\frac{2}{3}$ kg
(3) $\frac{1}{4}$ kg
(4) 2 kg

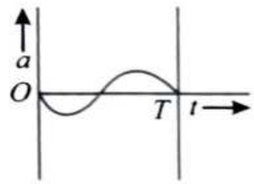
47. An electron of mass m and a photon have same energy E. The ratio of de-Broglie wavelengths associated with them is:
(c being velocity of light)
(1) $\left(\frac{E}{2m}\right)^{\frac{1}{2}}$
(2) $c(2mE)^{\frac{1}{2}}$
(3) $\frac{1}{c}\left(\frac{2m}{E}\right)^{\frac{1}{2}}$
(4) $\frac{1}{c}\left(\frac{E}{2m}\right)^{\frac{1}{2}}$

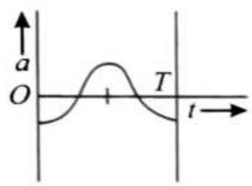
48. There is an air filled 1 pF parallel plate capacitor. When the plate separation is doubled and the space is filled with wax, the capacitance increases to 2 pF. The dielectric constant of wax is:
(1) 2
(2) 4
(3) 6
(4) 8

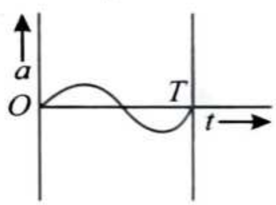
49. The S.I. unit of magnetic permeability is:
(1) Am^{-1}.
(2) A-m.
(3) Henry m^{-1}.
(4) no unit, it is a dimensionless number.

50. The oscillation of a body on a smooth horizontal surface is represented by the equation, $x = A \cos \omega t$
where, x = displacement at time t
ω = frequency of oscillation
Which one of the following graphs shows correctly the variation of a with t?
Here, a = acceleration at time t.
T = time period

(1) 

(2) 

(3) 

(4) 

Chemistry : Section-A (Q. No. 51 to 85)

51. Carbon and oxygen combine to form two oxides, carbon monoxide and carbon dioxide in which the ratio of the weights of carbon and oxygen is respectively 12:16 and 12 : 32. These figures illustrate the:
(1) Law of multiple proportions
(2) Avogadro's Law
(3) Law of conservation of mass
(4) Law of constant proportions

52. Which of the following is **incorrect** statement about an ionic compound?
(1) The higher the temperature, the more the solubility.
(2) The higher the dielectric constant of the solvent, the more the solubility.
(3) The higher the dipole moment of the solvent, the more the solubility.
(4) The higher the lattice energy, the more the solubility.

53. The **correct** formula of borax is:
(1) $Na_2[B_4O_5(OH)_4].8H_2O$
(2) $Na_2B_4O_7.4H_2O$
(3) $Na_2B_4O_7.8H_2O$
(4) $Na_2[B_4O_5(OH)_4].10H_2O$

54. Which of the following compounds will give silver mirror with ammonical silver nitrate?

A. Formic acid B. Formaldehyde
C. Benzaldehyde D. Acetone
Choose the correct answer from the options given below:
(1) pyramidal
(1) C and D only
(2) A, B and C only
(3) A only
(4) B and C only

55. The shape of ammonia molecule is:
(1) pyramidal
(2) octahedral
(3) tetrahedral
(4) trigonal planar

56. The metallic bond strength in first transition series increase from:
(1) Sc to Mn
(2) Sc to Cr
(3) Cr to Zn
(4) Sc to Cu

[Contd...

57. The degree of dissociation of pure water at 25 °C is found to be 1.8×10^{-9}. The dissociation constant of water (K_d) at 25 °C is:

(1) 10^{-14}
(2) 1.8×10^{-16}
(3) 5.56×10^{-13}
(4) 1.8×10^{-14}

58. Ligand in a complex salt are:

(1) anions linked by coordinate bonds to a central metal atom or ion.
(2) cations linked by coordinate bonds to a central metal atom or ion.
(3) molecules linked by ionic bonds to a central metal atom or ion.
(4) ions or molecule linked by coordinate bonds to a central atom or ion.

59. Match **List-I** with **List-II.**

	List-I (Coordination compound)		**List-II (Isomerism)**
(A)	$[Co(NH_3)_5(SO_4)]Br$ and $[Co(NH_3)_5Br]SO_4$	(I)	Coordination isomerism
(B)	$[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2.H_2O$	(II)	Linkage isomerism
(C)	$[Co(NH_3)_5(CN)Cl_2]$ and $[Co(NH_3)_5(NC)Cl_2]$	(III)	Ionisation isomerism
(D)	$[Cr(NH_3)_6][Co(CN)_6]$ and $[Co(NH_3)_6][Cr(CN)_6]$	(IV)	Solvate isomerism

Choose the **correct** answer from the options given below:

(1) A-III, B-IV, C-II, D-I
(2) A-III, B-II, C-IV, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-II, B-III, C-IV, D-I

60. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): CH_3NH_2 is more basic than $C_6H_5NH_2$.

Reason (R): As compared to aniline, methylamine is more basic due to +I effect of $-CH_3$ group.

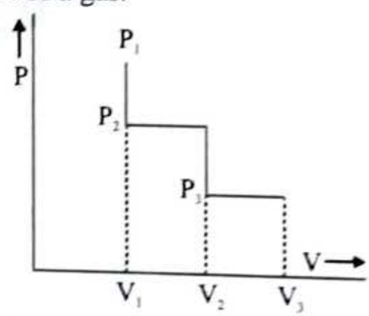
In the light of the above statements, choose the **correct** answer from the options given below:

(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

61. The relationship between standard reduction potential of a cell and equilibrium constant is:

(1) $E^o_{cell} = \dfrac{n}{0.059}\log K_C$
(2) $E^o_{cell} = \dfrac{0.059}{n}\log K_C$
(3) $E^o_{cell} = 0.059\, n \log K_C$
(4) $E^o_{cell} = \dfrac{\log K_C}{n}$

62. The following diagram represents of (P – V) changes of a gas.



Select isochoric change:

(1) When pressure changes from P_1 to P_2
(2) When pressure changes from P_2 to P_3
(3) When pressure changes from P_1 to P_3
(4) When pressure changes from P_3 to P_1

63. The pair having similar magnetic moment is:
(1) Ti^{3+}, V^{3+} (2) Cr^{3+}, Mn^{2+}
(3) Mn^{2+}, Fe^{3+} (4) Fe^{2+}, Mn^{2+}

64. Given below are two statements:
Statement-I: In the ionic equation,
$BiO_3^- + 6H^+ + xe^- \rightarrow Bi^{3+} + 3H_2O$
The values of x is 2.
Statement-II: In $MnO_4^- \rightarrow MnO_2$, transfer of five electrons takes place.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

65. The reddish brown gas that evolved in the preliminary test of nitrite ion is:
(1) NO (2) NO_2
(3) Br_2 (4) I_2

66. S_N2 reaction on a chiral center proceeds via:
(1) Complete Retention
(2) Complete Inversion
(3) Complete Racemization
(4) Partial Racemization

67. Match **List-I** with **List-II.**

	List-I (Elements)		**List-II (Their relations)**
(A)	Sc, Ca	(I)	Same period
(B)	Sn, Ge	(II)	Same group
(C)	Ag, Cr	(III)	Diagonal relationship
(D)	Be, Al	(IV)	Same block and different group

Choose the **correct** answer from the options given below:
(1) A-IV, B-III, C-II, D-I
(2) A-III, B-II, C-IV, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-I, B-II, C-IV, D-III

68. Which is **correct** statement for the given acids?
(1) Phosphinic acid is a diprotic acid while phosphonic acid is a monoprotic acid.
(2) Phosphinic acid is a monoprotic acid while phosphonic acid is a diprotic acid.
(3) Both are diprotic acid.
(4) Both are triprotic acids.

69. Splitting of spectral lines when atoms are subjected to strong electric field is called:
(1) Zeeman effect
(2) Stark effect
(3) Photoelectric effect
(4) None of these

70. Select the incorrect statement/s :
A. Alkanes contain carbon-carbon sigma (σ) bonds.
B. Magnitude of torsional strain depends upon the angle of rotation about C–C bond.
C. Staggered form of ethane has the maximum torsional strain and the eclipsed form has the least torsional strain.
D. Staggered conformation of ethane is less stable than the eclipsed conformation.
(1) A and B
(2) B, C and D
(3) C and D
(4) B and C

71. Given below are two statements:

Statement-I: All soluble sulphides gives white ppt. with $BaCl_2$ solution.

Statement-II: BaS is insoluble in water.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement-I is correct but Statement-II is incorrect.

(2) Statement-I is incorrect but Statement-II is correct.

(3) Both Statement-I and Statement-II are correct.

(4) Both Statement-I and Statement-II are incorrect.

72. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): Decomposition of potassium chlorate is an example of redox reaction.

Reason (R): There is no change in the oxidation state of potassium in decomposition of potassium chlorate.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) **A** is true but **R** is false.

(2) **A** is false but **R** is true.

(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.

(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

73. How many periods and vertical columns are there in the modern periodic table?

(1) 8, 12

(2) 6, 8

(3) 7, 18

(4) 6, 18

74. For a reversible reaction, the rate constants for the forward and backward reactions are 0.16 and 4×10^4 respectively. What is the value of equilibrium constant of the reaction?

(1) 0.25×10^6

(2) 2.5×10^5

(3) 4×10^{-6}

(4) 4×10^{-4}

75. Given below are two statements:

Statement-I: In a reaction $A(g) \rightleftharpoons B(g)$ at equilibrium, total number of moles in a closed system at new equilibrium is less than the old equilibrium if some amount of a substance is removed from a system.

Statement-II: The number of moles of the substance which is removed, is partially compensated as the system reached to new equilibrium.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

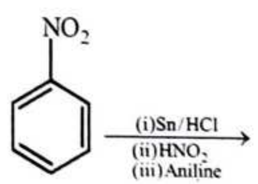
(1) Statement-I is correct but Statement-II is incorrect.

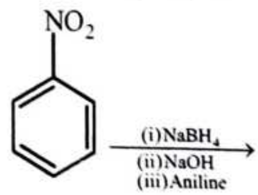
(2) Statement-I is incorrect but Statement-II is correct.

(3) Both Statement-I and Statement-II are correct.

(4) Both Statement-I and Statement-II are incorrect.

76. Which of the following reaction/s will not give p-aminoazobenzene?

(I)


(II)


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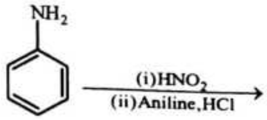
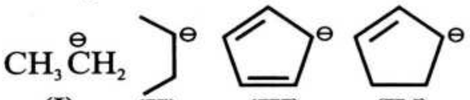
(III) 

(1) II only (2) I and II
(3) III only (4) I and III only

77. The –OH group of methyl alcohol can not be replaced by chlorine by the action of:
(1) chlorine.
(2) hydrogen chloride.
(3) phosphorous trichloride.
(4) phosphorous pentachloride.

78. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A): Temperature is an extensive property.
Reason (R): Internal energy is a state function.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

79. Given below are two statements:
Statement-I: The **correct** decreasing order of stability is: III > IV > I > II.



(I) (II) (III) (IV)

Statement-II: $\overset{\ominus}{C}H_3$ is less stable that $\overset{\ominus}{C}H_2 - NO_2$.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

80. Match **List-I** with **List-II.**

	List-I (Solution)		List-II (Solute in solvent)
(A)	Amalgum of mercury with sodium	(I)	Solid in gas
(B)	Chloroform mixed with nitrogen gas	(II)	Liquid in solid
(C)	Glucose dissolved in water	(III)	Liquid in gas
(D)	Camphor in nitrogen gas	(IV)	Solid in liquid

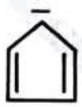
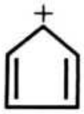
Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-IV, D-I
(2) A-III, B-II, C-IV, D-I
(3) A-II, B-I, C-III, D-IV
(4) A-IV, B-III, C-II, D-I

81. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A): Wurtz reaction is not preferred for the preparation of alkanes containing odd number of carbon atoms.
Reason (R): The reagent used in Wurtz reaction is Zn/dry ether.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

82. What is the oxidation state of Xe in Ba_2XeO_6?
(1) 0
(2) +4
(3) +6
(4) +8

83. Which of the following orbital has two spherical nodes?
(1) 2s
(2) 4s
(3) 3d
(4) 6f

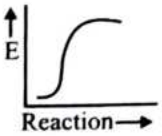
84. Given below are two statement: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A): Maltose is a reducing sugar.
Reason (R): The free aldehyde group can be produced at C-1 of second glucose unit in solution and it shows reducing properties.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

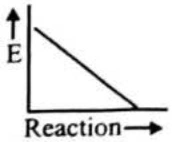
85. Which of the following compounds is **not** aromatic?
(1) 
(2) 
(3) 
(4) 

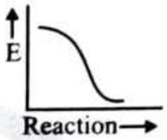
Chemistry : Section-B (Q. No. 86 to 100)

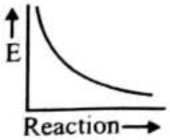
86. A compound (80 g) on analysis gave C = 24 g, H = 4 g, O = 32 g. its empirical formula is:
(1) $C_2H_2O_2$
(2) C_2H_2O
(3) CH_2O_2
(4) CH_2O

87. The Λ^{∞} of NH_4Cl, NaOH and NaCl are 129.8, 217.4 and 108.9 $ohm^{-1}\ cm^2\ eq^{-1}$ respectively. The Λ^{∞} of NH_4OH is ($S\ cm^2\ eq^{-1}$):
(1) 238.3
(2) 218
(3) 240
(4) 260

88. Which graph shows zero activation energy?
(1) 

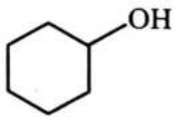
(2) 

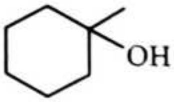
(3) 

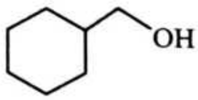
(4) 

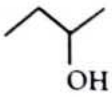
89. Among the following statements:
A. If the salt bridge is removed from Galvanic cell, then voltage drops to zero.
B. The half-cell in which oxidation takes place is called anode.
C. The conductivity of electrolytic (ionic) solutions depends on the nature of the electrolyte.
The **correct** statements is/are;
(1) A only
(2) A and B
(3) A, B and C
(4) A and C

90. Zero dipole moment is possessed by:
(1) PCl_3
(2) BF_3
(3) ClF_3
(4) NH_3

91. Which alcohol will react fastest with HCl and anhy. $ZnCl_2$?
(1) 

(2) 

(3) 

(4) 

92. Which one of the following is **not** correct for an ideal solution?

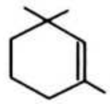
(1) It must obey Raoult's law

(2) $\Delta H_{mix} = 0$

(3) $\Delta V_{mix} = 0$

(4) $\Delta H_{mix} \neq 0, \Delta V_{mix} \neq 0$

93. Give the IUPAC name of the compound:



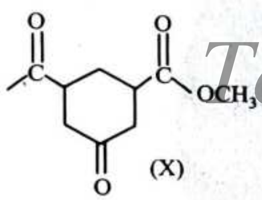
(1) 1, 1, 3-Trimethylcyclohex-2-ene

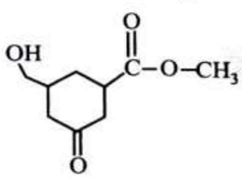
(2) 1, 3, 3-Trimethylcyclohex-1-ene

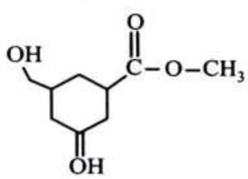
(3) 1, 1, 5-Trimethylcyclohex-5-ene

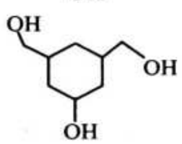
(4) 2, 6, 6-Trmethylcyclohex-1-ene

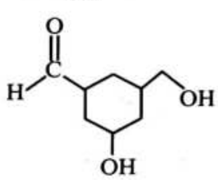
94. Predict the product when given compound (X) reacts with $LiAlH_4$:



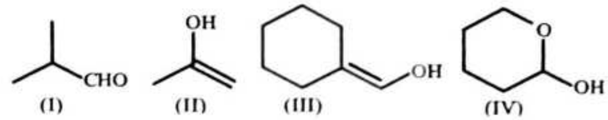
(1) 

(2) 

(3) 

(4) 

95.



Which among the above compound/s does / do not form silver mirror when treated with Tollen's reagent?

(1) Only (IV) (2) (III) and (IV) only

(3) Only (II) (4) (I), (III) and (IV) only

96. A first order reaction takes 40 min for 30% decomposition. Calculate $t_{1/2}$:

(Given log 3 = 0.693, log 7 = 0.845)

(1) 77.7 min (2) 27.2 min

(3) 55.3 min (4) 67.3 min

97. Which of the following ions is coloured in solution?

(1) Zn^{2+} (2) Ti^{4+}

(3) Cu^{+} (4) V^{3+}

98. o-nitrophenol is steam volatile whereas para-nitrophenol is **not** due to;

(1) Intramolecular H-bonding present in ortho-nitrophenol

(2) Intermolecular H-bonding

(3) Intramolecular H-bonding present in para nitrophenol

(4) None of these

99. Specific conductance of 0.1 M nitric acid is 6.3×10^{-2} ohm^{-1} cm^{-1}. The molar conductance of solution:

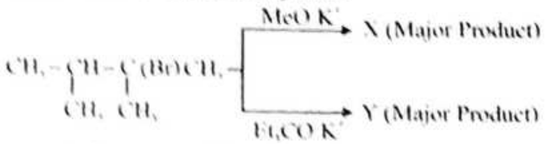
(1) 630 ohm^{-1} cm^2 $mole^{-1}$

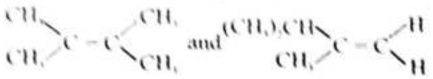
(2) 315 ohm^{-1} cm^2 $mole^{-1}$

(3) 100 ohm^{-1} cm^2 $mole^{-1}$

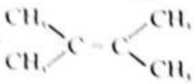
(4) 6300 ohm^{-1} cm^2 $mole^{-1}$

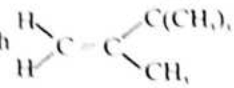
100. X and Y are respectively are:



(1) 

(2) 

(3) Both 

(4) Both 

Botany : Section-A (Q. No. 101 to 135)

101. The cell organelle which is double membrane bound, site of aerobic respiration, and produces cellular energy in the form of ATP is:

(1) Chloroplast.

(2) Ribosomes.

(3) Mitochondria.

(4) E.R.

102. Choose the **incorrect** statement for the universal rules of biological nomenclature:

(1) Biological names are generally in Latin or latinised.

(2) The first word in a biological name represents the specific epithet while the second component denotes the genus.

(3) Both the words in a biological name, when handwritten, are separately underlined, or printed in italics to indicate their Latin origin.

(4) It was proposed by Carolus Linnaeus.

103. The group of PGRs that are involved in growth promoting activities, such as cell division, cell enlargement, pattern formation, tropic growth, flowering, fruiting and seed formation are;

(1) gibberellins, ethylene and abscisic acid.

(2) auxins, gibberellins and cytokinins.

(3) abscisic acid, cytokinin only.

(4) ethylene and abscisic acid only.

104. Who used recombination frequencies to map gene positions on chromosomes?

(1) Gregor Mendel

(2) Alfred Sturtevant

(3) Walter Sutton and Theodore Boveri

(4) Thomas Hunt Morgan

105. Which events occur during the M phase of the cell cycle?

(a) DNA replication

(b) Chromosome condensation

(c) Nuclear envelope breakdown

(d) Spindle formation

(e) Centrosome duplication

Choose the correct options:

(1) (a), (b) and (e) only

(2) (b), (c) and (d) only

(3) (c), (d) and (e) only

(4) (a), (c) and (e) only

106. Which of the following bacteria are known for nitrogen fixation while being free-living in soil?

(a) *Rhizobium*	(b) *Mycobacterium*
(c) *Azospirillum*	(d) *Azotobacter*
(e) *Lactobacillus*	(f) *Streptococcus*
(1) (a), (c) and (f)	(2) (b), (c) and (d)
(3) (c) and (d)	(4) (a), (b) and (e)

107. Alternate type of phyllotaxy is present in all, the following, **except**;

(1) china rose.

(2) mustard.

(3) sun flower.

(4) guava.

[Contd...

108. Which of the following substances are produced by yeast?

(a) Citric acid (b) Ethanol
(c) Statins (d) Lactic acid
(e) Proteases

Options:

(1) (a), (b) and (c) only
(2) (b) and (c) only
(3) (b), (c) and (e) only
(4) (b) and (e) only

109. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): Cleistogamous flowers are invariably autogamous.

Reason (R): In cleistogamous flowers, there is no chance of cross-pollen landing on the stigma.

In the light of above statements, choose the **correct** answer from the options given below:

(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

110. Identify the **incorrect** statement w.r.t DFC.

(1) The detritus food chain (DFC) begins with dead organic matter.
(2) It is made up of decomposers which are autotrophic organisms, mainly fungi and bacteria.
(3) Decomposers secrete digestive enzymes that breakdown dead and waste materials into simple, inorganic materials, which are subsequently absorbed by them.
(4) The decomposers are also known as saprotrophs.

111. Identify the **incorrect** statements from the following.

A. The hard outer layer of pollen grain is called the intine.
B. The inner wall of the pollen grain is called the exine.
C. Pollen grain exine has prominent apertures called germ pores where sporopollenin is absent.
D. Intine of pollen grain is a thin and continuous layer.
E. The cytoplasm of pollen grain is surrounded by a plasma membrane.

Choose the *most appropriate* answer from the options given below:

(1) A and E (2) B and D
(3) A and B (4) C and D

112. The artificial classification system is mainly based on which of the following characteristics?

(1) Vegetative characters or androecium structure
(2) Ultrastructure, anatomy, and embryology
(3) External and internal features
(4) Chromosome number

113. State **true (T)** or **false (F)** for the following statements.

A. The biomass of a species is expressed in terms of fresh or dry weight.
B. Only 1.0 per cent of the energy is transferred to each trophic level from the lower trophic level.
C. Organisms at each trophic level depend on those at the lower trophic level for their energy demands.
D. Producers belong to the first trophic level, herbivores and carnivores to the second trophic level.

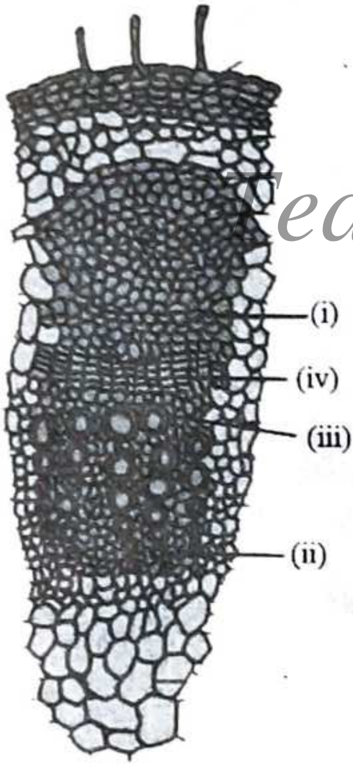
Choose the most appropriate answer from the options given below:

	A	B	C	D
(1)	T	F	T	F
(2)	F	T	T	T
(3)	T	T	F	F
(4)	F	F	F	T

114. A DNA nucleotide chain has AGCTTCGA sequence. The nucleotide sequence of other chain would be;
(1) TCGAAGCT. (2) GCTAAGCT.
(3) TAGCATAT. (4) GATCCTAG.

115. The wall layers which perform the function of protection of microsporangia are;
(1) tapetum and middle layer only.
(2) epidermis, endothecium and tapetum.
(3) endothecium, epidermis and middle layer.
(4) epidermis and middle layer only.

116. Identify the **correct** labels for the given figure of dicot stem.



(1) **(i)** Protoxylem **(ii)** Cambium **(iii)** Phloem **(iv)** Metaxylem
(2) **(i)** Phloem **(ii)** Protoxylem **(iii)** Metaxylem **(iv)** Cambium
(3) **(i)** Phloem **(ii)** Cambium **(iii)** Protoxylem **(iv)** Metaxylem
(4) **(i)** Metaxylem **(ii)** Cambium **(iii)** Phloem **(iv)** Protoxylem

117. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Pulvinus	(I)	Swollen leaf base
(B)	Root cap	(II)	Green expanded part of leaf
(C)	Lamina	(III)	Thimble like structure
(D)	Petiole	(IV)	Stalk that attaches the leaf blade to stem

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-II, D-IV
(2) A-III, B-II, C-I, D-IV
(3) A-I, B-III, C-II, D-IV
(4) A-II, B-I, C-III, D-IV

118. Select the **incorrect** statement.
(1) The ratio of the volume of CO_2 evolved to the volume of O_2 consumed in respiration is called the respiratory quotient (RQ) or respiratory ratio.
(2) The respiratory quotient depends upon the type of respiratory substrate used during respiration.
(3) Pure proteins or fats and carbohydrates are always used as respiratory substrates.
(4) When carbohydrates are used as substrate, the RQ will be 1.

119. Mesosome which is formed by the extensions of plasma membranes into the cell does **not** help in:
(1) Formation of cell wall
(2) Replication of DNA
(3) Respiration
(4) Storing pigments

[Contd...

120. Given below are two statements:

Statement-I: Micro-organisms such as *Lactobacillus* and others commonly called lactic acid bacteria grow in milk and convert it to curd.

Statement-II: During growth, the LAB produce enzymes that coagulate and partially digest the milk proteins.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

121. Identify the **correct** statements from the following:

A. In case of plants, classes with a few similar characters are aligned to a higher category called phylum.

B. Sub-categories have also been developed in the taxonomic hierarchy to facilitate more sound and scientific placement of various taxa.

C. Class includes related orders.

D. Convolvulaceae family is included in order Polymoniales on the basis of its floral characters.

Choose the **correct** answer from the options given below:

(1) A and D
(2) A and B
(3) B, C and D
(4) All of these

122. Given below are two statement: one is labeled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): A filament in a flower consists of three parts namely stigma, style and ovary.

Reason (R): The stigma is present at the tip of the style and is the receptive surface for pollen grains.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

123. The Severo Ochoa enzyme is;

(1) DNA polymerase.
(2) reverse transcriptase.
(3) polynucleotide phosphorylase.
(4) DNA-dependent DNA polymerases.

124. Identify the **incorrect** statement w.r.t the factors affecting photosynthesis.

(1) Increase in incident light beyond a point causes the breakdown of chlorophyll and a decrease in photosynthesis.
(2) Carbon dioxide is the major limiting factor for photosynthesis.
(3) The C_4 plants respond to higher temperatures and show higher rate of photosynthesis while C_3 plants have a much lower temperature optimum.
(4) The dark reactions being enzymatic are temperature independent.

125. Cells are divided and new cells are formed from pre-existing cells (*Omnis cellula-e-cellula*) was first explained by:

(1) Theodore Schwann.
(2) Rudolf Virchow.

[Contd...

(3) Matthias Schleiden.
(4) All of these.

126. Identify the **correct** statements w.r.t sporophyte of mosses.
A. Is more elaborate than that in liverworts.
B. Consists of foot, seta and capsule.
C. Spores are present in capsule.
D. Spores are formed after meiosis.
E. Elaborate mechanism of spore dispersal is present in them.
Choose the **correct** answer from the options given below:
(1) All are correct (2) Only A is correct
(3) Only B is correct (4) Only C is correct

127. In which of the given organisms, Taylor et al. have proved semiconservative mode of chromosome replication by using radioactive-thymidine?
(1) *Vicia faba* (2) *E.coli*
(3) *Pneumococcus* (4) *Streptococcus*

128. Identify the **incorrect** statement w.r.t pollination.
(1) Plants use abiotic (animals) and biotic (wind and water) agents to achieve pollination.
(2) Pollination by water is quite rare in flowering plants.
(3) Majority of plants use biotic agents for pollination.
(4) Wind pollination is quite common in grasses.

129. Given below table shows a hypothetical population of a country from 2002-2005.

S.No	Year	Birth rate (per 1000 individuals)	Death rate (per 1000 individuals)
(a)	2005	16.2	7.3
(b)	2004	16.7	7.6
(c)	2003	17.0	7.5
(d)	2002	17.2	7.4

Arrange the following years in order of decreasing natural increase rate (r) of the population in a country.
Options:
(1) (a), (b), (c), (d) (2) (d), (c), (b), (a)
(3) (b), (a), (c), (d) (4) (c), (b), (a), (d)

130. Identify the number of plants that have perigynous flowers and choose the **correct** option:

Peach, plum, mustard, brinjal, ray florets of sunflower

(1) Four (2) Three
(3) Five (4) Two

131. Which of the following contain 70S ribosomes?
(a) Mitochondria (b) Chloroplasts
(c) Lysosomes (d) Cytoplasm
(e) Bacteria
Choose the correct answer from the options given below:
(1) (a) only
(2) (a) and (b) only
(3) (a), (b), and (e) only
(4) (b), (c), and (d) only

132. Identify the **correct** sequence of microsporogenesis.
(1) Microspore tetrad → Pollen mother cell (PMC) → Sporogenous tissue
(2) Sporogenous tissue → Microspore tetrad → PMC
(3) PMC → Microspore tetrad → Sporogenous tissue
(4) Sporogenous tissue → PMC → Microspore tetrad

133. Select the **correct** statements from the following w.r.t transcription.
A. The DNA-dependent RNA polymerase catalyse the polymerisation in only one direction.
B. The coding sequences or expressed sequences are defined as introns.

[Contd...

C. Intervening sequences do appear in processed RNA.

D. In bacteria, translation can begin much before the mRNA is fully transcribed.

(1) A and D

(2) A and C

(3) B and C

(4) B and D

134. Choose the **incorrectly** matched pair w.r.t to the condition of stamens:

(1) Epipetalous - Brinjal

(2) Monoadelphous - China rose

(3) Epiphyllous - Pea

(4) Polyadelphous – *Citrus*

135. Arrange the following in the **correct** sequence of usage w.r.t STP (Sewage Treatment Plant):

A. Anaerobic sludge digester

B. Large aeration tank

C. Primary settling tank

(1) C → B → A

(2) A → B → C

(3) B → C → A

(4) C → A → B

Botany : Section-B (Q. No. 136 to 150)

136. Recombination nodules on homologous chromosomes during meiosis are;

(1) Formed in the final stage of prophase I.

(2) Formed in meiosis II.

(3) Visible as Y shaped structures.

(4) The sites at which crossing over occurs.

137. Match **List-I** with **List-II**.

	List-I		**List-II**
(A)	Phaeophyceae	(I)	Pyrenoids
(B)	Mosses	(II)	Pyriform gametes
(C)	Chlorophyceae	(III)	Gemma cups
(D)	Liverworts	(IV)	Protonema

Choose the correct answer from the options given below:

(1) A-II; B-III; C-IV; D-I

(2) A-III; B-I; C-IV; D-II

(3) A-IV; B-III; C-II; D-I

(4) A-II; B-IV; C-I; D-III

138. Choose the **incorrect** statement:

(1) Meiosis involves two sequential cycles of nuclear and cell division called meiosis I and meiosis II but only a single cycle of DNA replication.

(2) Meiosis I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the S phase.

(3) Meiosis involves pairing of homologous chromosomes and recombination between sister chromatids of homologous chromosomes.

(4) Four haploid cells are formed at the end of meiosis II.

139. Choose the **incorrect** statement:

(1) *Chlorella* is a unicellular alga rich in protein.

(2) *Chlorella* is used as a food supplement by space travellers.

(3) The product obtained from *Gracilaria* is used to grow microbes.

(4) *Laminaria* and *Sargassum* are used in preparations of ice-creams and jellies.

140. Some interactions are given in the table, mark the option with the **correct** labelling.

Species A	**Species B**	**Name of Interaction**
+	–	A
+	0	B
–	0	C

(1) A-Parasitism, B-Commensalism, C-Amensalism

(2) A-Commensalism, B-Parasitism, C-Amensalism

(3) A-Parasitism, B-Amensalism, C-Commensalism

(4) A-Amensalism, B-Commensalism, C-Parasitism

141. Choose the **incorrectly** matched pair.

(1)	Tracheids	-	one of the main water transporting elements.
(2)	Vessel	-	characteristic feature of angiosperms.
(3)	Xylem fibres	-	highly thickened walls.
(4)	Xylem parenchyma	-	provide mechanical support.

142. Photorespiration is a wasteful process because there is;

(1) neither synthesis of phosphoglycerate, nor of ATP.
(2) neither synthesis of phosphoglycerate, nor of phosphoglycolate.
(3) neither synthesis of sugars, nor of ATP.
(4) neither synthesis of phosphoglycolate nor of ATP.

143. Identify the **mismatched** pair.

(1)	Gemmae	-	asexual buds
(2)	Coralloid roots	-	N_2 fixation
(3)	Mycorrhiza	-	symbiotic association of fungi with roots of lower plants
(4)	Prothallus	-	free living thalloid gametophyte

144. Referring to the pictures given below, choose the option having **correct** sequence of events w.r.t meiosis I.

(A) (B) (C)

(1) B → A → C
(2) A → B → C
(3) C → A → B
(4) A → C → B

145. Given below are two statements:

Statement-I: Ascomycetes are commonly known as sac-fungi.

Statement-II: In ascomycetes asexual spores are conidia produced exogenously on the special mycelium called conidiophores.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

146. Select the **incorrect** statement.

(1) The 'bakane' (foolish seedling) disease of rice seedlings, was caused by a fungal pathogen *Gibberella fujikuroi*. The active substances were later identified as auxin.
(2) Auxins control xylem differentiation and help in cell division.
(3) Miller et al. (1955) identified and crystallised the cytokinesis promoting active substance that they termed kinetin.
(4) Cousins confirmed the release of a volatile substance from ripened oranges that hastened the ripening of stored unripened bananas. Later it was identified as ethylene, a gaseous PGR.

147. Match **List-I** with **List-II.**

	List-I		**List-II**
(A)	G_1 Phase	(I)	Cell grows and organelle duplication
(B)	S Phase	(II)	DNA replication and chromosomes duplication

[Contd...

(C)	G_2 Phase	(III)	Cytoplasmic growth
(D)	M- Phase	(IV)	Alignment of chromosomes on equatorial plate

Choose the **correct** answer from the options given below:

(1) A-II, B-IV, C-I, D-III
(2) A-I, B-II, C-III, D-IV
(3) A-III, B-II, C-IV, D-I
(4) A-IV, B-III, C-II, D-I

148. In the C_4 plants, Calvin pathway occurs in;
(1) mesophyll cells.
(2) bundle sheath cells.
(3) both mesophyll and bundle sheath cells.
(4) none of these.

149. When a host fish species becomes extinct, its unique assemblage of parasites also meets the same fate. This Statement-Is **true** about;
(1) alien species invasions.
(2) co-extinctions.
(3) over-exploitation.
(4) habitat loss and fragmentation.

150. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A): Growth, differentiation and development are very closely related events in the life of a plant.
Reason (R): Development in plants (i.e., both growth and differentiation) is under the control of intrinsic and extrinsic factors.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

Zoology : Section-A (Q. No. 151 to 185)

151. Select the **correct** sequence of embryonic development.
(1) Morula → Zygote → Blastocyst
(2) Blastocyst → Morula → Zygote
(3) Zygote → Morula → Blastocyst
(4) Morula → Blastocyst → Zygote

152. Select the **correct** statements.
A. Platyhelminthes are triploblastic, pseudocoelomate and bilaterally symmetrical organisms.
B. Ctenophores reproduce only sexually and fertilisation is external.
C. In tapeworm, fertilisation is internal but sexes are not separate.
D. Ctenophores are exclusively marine, diploblastic and bioluminescent organisms.
E. In sponges, fertilisation is external and development is direct.
Choose the **correct** answer from the options given below:
(1) A, C and D only
(2) B, C and D only
(3) A and E only
(4) B and D only

153. What triggers activation of protoxin to active Bt toxin of *Bacillus thuringiensis* in boll worm?
(1) Body temperature
(2) Moist surface of midgut
(3) Alkaline pH of gut
(4) Acidic pH of stomach

154. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A): Enzymes isolated from thermophilic organisms are thermally stable.
Reason (R): They retain their catalytic power even at higher temperature.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

155. What is the main function of parathormone (PTH)?
(1) Regulate blood calcium levels.
(2) Regulate metabolism.
(3) Stimulate insulin secretion.
(4) Control reproductive cycles.

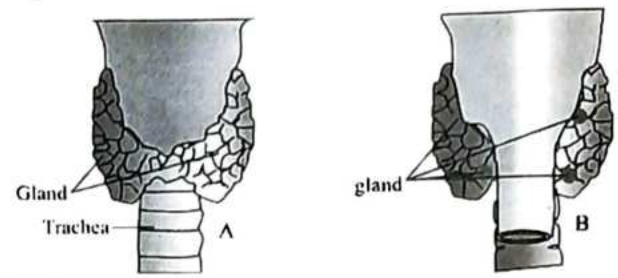
156. Given below are two statements:
Statement-I: Human heart is an endodermally derived organ.
Statement-II: Tricuspid and mitral valves are open during joint diastole.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

157. Match the **List-I** and **List-II**:

	List-I		**List-II**
A.	Widal Test	I.	*Wuchereria*
B.	Filariasis	II.	Carcinogens
C.	Diagnosis of cancer	III.	Typhoid fever
D.	X-rays	IV.	Biopsy

Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-IV, D-III
(2) A-II, B-I, C-IV, D-III
(3) A-III, B-IV, C-I, D-II
(4) A-III, B-I, C-IV, D-II

158. Below are two pictures showing different views of trachea and glands around it. Which of the following choices **correctly** labels the view and the gland that could be seen in the view for both the pictures?



(1) **A**-Dorsal-Thymus, **B**-Ventral-Parathyroid
(2) **A**-Dorsal-parathyroid, **B**-Ventral-Thymus
(3) **A**-Ventral-Thyroid, **B**-Dorsal-Parathyroid
(4) **A**-Ventral- Parathyroid, **B**-Dorsal-Thyroid

159. Match the **List-I** and **List-II**:

	List-I		**List-II**
A.	Conventional method of diagnosis	I.	Bone marrow transplantation
B.	Method of early diagnosis	II.	Serum analysis
C.	Treatment of emphysema	III.	ELISA
D.	Gene therapy	IV.	α-1 antitrypsin

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-IV, D-I
(2) A-II, B-I, C-IV, D-III
(3) A-IV, B-I, C-III, D-II
(4) A-III, B-IV, C-I, D-II

160. Given below are two statements:
Statement-I: Capacity to generate a whole plant from any cell/explant is called totipotency.
Statement-II: In 1995, an American company got patent rights on Basmati rice through the US Patent and Trademark Office.

[Contd...

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

161. Choose the option which have all **correctly** matched sets.

(A)	Heart	Four chambered
(B)	Sino-atrial node	Right upper corner of the right atrium
(C)	Pericardium	Walls of arteries and veins
(D)	QRS complex	Contraction of atria

(1) (A) and (C) only (2) (A) and (B) only
(3) (B) and (D) only (4) (C) and (D) only

162. Saheli was developed by:

(1) AIIMS. (2) NDRI.
(3) CDRI. (4) WHO.

163. Match the **List-I** and **List-II**:

	List-I		List-II
A.	Pneumotaxic Centre	I.	Alveoli
B.	O_2 dissociation curve	II.	Pons region of brain
C.	Carbonic anhydrase	III.	Haemoglobin
D.	Primary site of exchanges	IV.	RBCs

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-IV, D-I
(2) A-III, B-II, C-IV, D-I
(3) A-IV, B-I, C-III, D-II
(4) A-I, B-III, C-IV, D-II

164. Which of the following is/are **correct** regarding vasectomy?

(1) No sperms are found in semen.
(2) Vas deferens tied and cut.
(3) It does not affect spermatogenesis.
(4) All of these.

165. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):

Assertion (A): Heroin is chemically diacetylmorphine.

Reason (R): It is obtained by acetylation of cocaine.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

166. Dinosaurs suddenly disappeared from earth about:

(1) 200 mya.
(2) 65 mya.
(3) 15 mya.
(4) 320 mya.

167. Match the **List-I** and **List-II:**

	List-I		List-II
A.	Sponges	I.	Lungs
B.	Earthworms	II.	Body surfaces
C.	Mollusca	III.	Moist cuticle
D.	Birds	IV.	Gills

Choose the **correct** answer from the options given below:

(1) A-II, B-III, C-IV, D-I
(2) A-II, B-IV, C-III, D-I

[Contd...

(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-III, C-I, D-II

168. Choose the **correct** match.
(1) Louis Pasteur – Theory of use and disuse
(2) Neanderthal man – Brain capacity 1400cc
(3) *Ichthyosaurs* – Fish like amphibians
(4) *Dryopithecus* – Man-like ape

169. Ringworm is caused by:
A. *Microsporum*.
B. *Trichophyton*.
C. *Ascaris*.
D. *Plasmodium*.
(1) A and B only (2) B and C only
(3) A, C and D only (4) All of these

170. Given below are two statements:
Statement-I: Proteins are polypeptides.
Statement-II: A nucleotide has two chemically distinct components.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

171. All of the following are examples of medicated IUDs, **except**:
(1) Multiload-375.
(2) progestasert.
(3) Lippes loop.
(4) LNG-20.

172. Choose the **incorrect** statement.
(1) Darwinian variations are large and non-directional.
(2) Vertebrate hearts and brains exhibit divergent evolution.
(3) Alfred Wallace worked in Malay Archipelago.
(4) Similar habitat results in selection of similar adaptive features in different organisms.

173. Select the **correct** statements.
A. When any plane passing through the central axis of the body divides the organism into two identical halves, it is called radial symmetry.
B. The body cavity, which is lined by mesoderm is called coelom.
C. Choanocytes or collar cells line the spongocoel and the canals.
D. Cnidarians exhibit two basic body forms called polyp and medusa.
E. Nephridia help in osmoregulation and excretion.
Choose the **correct** answer from the options given below:
(1) A, C and D only
(2) B, C and D only
(3) A and E only
(4) A, B, C, D and E only

174. Match the **List-I** and **List-II**:

	List-I		List-II
A.	Gregarious pest	I.	*Asterias*
B.	Adult with radial symmetry and larva with bilateral symmetry	II.	Scorpion
C.	Book lungs	III.	*Ctenoplana*
D.	Bioluminescence	IV.	Locust

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-II, D-III
(2) A-III, B-II, C-I, D-IV

(3) A-II, B-I, C-III, D-IV
(4) A-I, B-III, C-II, D-IV

175. Given below are two statements:
Statement-I: Galaxies contain stars and clouds of gas and dust.
Statement-II: The fitness, according to Darwin, refers ultimately and only to reproductive fitness.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

176. Select the **incorrect** statements.
A. Biotechnology deals with techniques of using live organisms or enzymes from organisms to produce products and processes useful to humans.
B. In the year 1863, the two enzymes responsible for restricting the growth of bacteriophage in *Escherichia coli* were isolated.
C. DNA is a negatively charged molecule.
D. A stirred-tank reactor is usually rectangular or with a curved base to facilitate the mixing of the reactor contents.
E. Retroviruses in animals have the ability to transform normal cells into cancerous cells.
Choose the **correct** answer from the options given below:
(1) A, C and D only
(2) B, C and D only
(3) A and E only
(4) B and D only

177. First discovered restriction endonuclease that always cuts DNA molecule at a particular point by recognising a specific sequence of six base pairs is:
(1) adenosine deaminase.
(2) thermostable DNA polymerase.
(3) *Hind* II.
(4) *EcoRI*.

178. Match the **List-I** and **List-II** w.r.t. cockroach:

	List-I		List-II
A.	Anal cerci	I.	7^{th} in females
B.	Boat-shaped sternum	II.	9^{th} segment in males
C.	Anal styles	III.	10^{th} segment
D.	Genital pouch	IV.	Bounded by 9^{th} and 10^{th} terga in males

Choose the **correct** answer from the options given below:
(1) A-IV, B-II, C-I, D-III
(2) A-II, B-III, C-IV, D-I
(3) A-I, B-IV, C-III, D-II
(4) A-III, B-I, C-II, D-IV

179. Find the **mismatch** pair.
(1) Streaming of protoplasm - Helps in locomotion in *Amoeba*
(2) Sarcoplasmic reticulum - Store house of Ca^{++} ions
(3) Knee joint - Hinge joint
(4) Clavicle and scapula - Pectoral girdle

180. In adults, each testis is oval in shape, with a length of about __**A**__ and a width of about __**B**__.
Choose the **correct** option to fill the blanks.
(1) **A** – 4 to 5 cm; **B** – 2 to 3 cm
(2) **A** – 2 to 3 cm; **B** – 4 to 5 cm
(3) **A** – 4 to 6 cm; **B** – 3 to 4 cm
(4) **A** – 2 to 4 cm; **B** – 2 to 3 cm

181. Match the **List-I** and **List-II**:

	List-I		List-II
A.	Tarsals	I.	14
B.	Phalanges	II.	1
C.	Metatarsals	III.	7
D.	Femur	IV.	5

Choose the **correct** answer from the options given below:

(1) A-III, B-I, C-IV, D-II
(2) A-I, B-II, C-III, D-IV
(3) A-II, B-III, C-IV, D-I
(4) A-IV, B-I, C-III, D-II

182. After implantation, the inner cell mass differentiates into outer __I__, middle __II__ and inner __III__.
Choose the option which fill the blanks **correctly**.
(1) mesoderm, endoderm, ectoderm
(2) endoderm, mesoderm, ectoderm
(3) ectoderm, mesoderm, endoderm
(4) mesoderm, ectoderm, endoderm

183. Which of the following is an example of adaptive radiation?
(1) The development of antibiotic resistance in bacteria.
(2) The diversification of Darwin's finches on the Galápagos Islands.
(3) The extinction of the dinosaurs.
(4) All of these

184. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A): Adenohypophysis consists of two portions, pars distalis and pars intermedia in humans.
Reason (R): The pars intermedia region of the pituitary is commonly called anterior pituitary.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) **A** is true but **R** is false.
(2) **A** is false but **R** is true.
(3) Both **A** and **R** are true and **R** is the correct explanation of **A**.
(4) Both **A** and **R** are true but **R** is NOT the correct explanation of **A**.

185. Match the **List-I** and **List-II:**

List-I (Type of Joint)		**List-II**	
A.	Cartilaginous Joint	I.	Between flat skull bones
B.	Ball and Socket Joint	II.	Between adjacent vertebrae in vertebral column
C.	Fibrous Joint	III.	Between carpal and metacarpal of thumb
D.	Saddle Joint	IV.	Between Humerus and Pectoral girdle

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-I, B-IV, C-III D-II
(3) A-II, B-IV, C-III, D-I
(4) A-III, B-I, C-II, D-IV

Zoology : Section-B (Q. No. 186 to 200)

186. Read the following statements.
A. Kidneys are situated between the levels of last thoracic and third lumbar vertebra.
B. On an average, 11000-12000 ml of blood is filtered by the kidneys per minute.
C. Glomerulus is a tuft of capillaries formed by the afferent arteriole.
D. The Henle's loop is hairpin-shaped and consists of a descending and ascending limb.
E. JGA is a special sensitive region formed by cellular modifications in the proximal convoluted tubule.
Choose the option with the **correct** statements from the options given below:
(1) A, C and D only
(2) B, C and D only
(3) A and E only
(4) B and D only

187. Match the **List-I** and **List-II**:

	List-I		List-II
A.	Emphysema	I.	2500 mL to 3000 mL
B.	Inspiratory Reserve Volume	II.	Reduce the duration of inspiration.
C.	Diaphragm	III.	Chronic disorder
D.	Pneumotaxic centre	IV.	Increase/decrease thoracic volume in antero-posterior axis.

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-I, D-IV
(2) A-III, B-I, C-IV, D-II
(3) A-II, B-IV, C-I, D-III
(4) A-IV, B-III, C-I, D-II

188. The edges of the infundibulum possess finger-like projections called:

(1) fimbriae. (2) cilia.
(3) microvilli. (4) All of these

189. Match the **List-I** and **List-II:**

	List-I		List-II
A.	Eosinophils	I.	6-8%
B.	Lymphocytes	II.	2-3%
C.	Neutrophils	III.	20-25%
D.	Monocytes	IV.	60-65%

Choose the **correct** answer from the options given below:

(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-I, C-III, D-II
(3) A-II, B-III, C-IV, D-I
(4) A-II, B-III, C-I, D-IV

190. Brain can be divided into:

(1) cerebrum, thalamus and hypothalamus.
(2) piamater, arachnoid and duramater.
(3) pons, cerebellum and medulla.
(4) forebrain, midbrain and hindbrain.

191. Match the **List-I** with **List-II**:

	List-I		List-II
A.	RNAi	I.	Cotton bollworms
B.	*cry I Ab*	II.	Corn borer
C.	*cry II Ab*	III.	Cellular defence
D.	Vector	IV.	*Agrobacterium*

Choose the **correct** answer from the options given below:

(1) A-III, B-II, C-I, D-IV
(2) A-I, B-II, C-IV, D-III
(3) A-I, B-II, C-III, D-IV
(4) A-III, B-IV, C-I, D-II

192. Given below are two statements:

Statement-I: A mature female frog can lay 500-1000 ova at a time.

Statement-II: In some countries, muscular legs of frog are used as food by man.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

193. Match the **List-I** and **List-II:**

	List-I		List-II
A.	CCK	I.	Kidney
B.	GIP	II.	Heart
C.	ANF	III.	Gastric gland
D.	ADH	IV.	Pancreas

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-III, D-I
(2) A-IV, B-III, C-II, D-I
(3) A-III, B-II, C-IV, D-I
(4) A-II, B-IV, C-I, D-III

194. Which of the following statements are **correct**?

A. During swallowing glottis can be covered by epiglottis.
B. Each terminal bronchiole gives rise to a thick regular walled vascularised bag like structure called alveoli.
C. Humans have two lungs.

[Contd...

D. Human lungs are covered by a single layered pleura.
E. The conducting part is also known as exchange part in human respiratory system.

Choose the most appropriate answer from the options given below:
(1) A, B and C only
(2) A, B, C, D and E
(3) A and B only
(4) A and C only

195. Match the **List-I** and **List-II**:

	List-I		List-II
A.	Platyhelminthes	I.	Cylindrical body with no segmentation.
B.	Echinoderms	II.	Warm blooded animals with direction development.
C.	Hemichordates	III.	Bilateral symmetry with incomplete digestive system.
D.	Aves	IV.	Endoskeleton made up of calcareous ossicles.

Choose the **correct** answer from the options given below:
(1) A-II, B-III, C-IV, D-I
(2) A-IV, B-I, C-II, D-III
(3) A-I, B-II, C-III, D-IV
(4) A-III, B-IV, C-I, D-II

196. Full form of MALT is:
(1) Macrophages Associated Lymphoid Tissue
(2) Macrophages Associated Lymphoid Tumor
(3) Mucosa Associated Lymphoid Tumor
(4) Mucosa Associated Lymphoid Tissue

197. Given below are two statements:
Statement-I: According to the 2011 census report, the population growth rate was less than 2 per cent, i.e., 20/1000/year.
Statement-II: Indian population crossed 1.2 billion in May 2011.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Statement-I is correct but Statement-II is incorrect.
(2) Statement-I is incorrect but Statement-II is correct.
(3) Both Statement-I and Statement-II are correct.
(4) Both Statement-I and Statement-II are incorrect.

198. At which stage of life the oogenesis process is initiated?
(1) Adult
(2) Puberty
(3) Embryonic development stage
(4) Birth

199. Brainstem of human brain consists of:
(1) mid-brain, pons and medulla oblongata.
(2) forebrain, cerebellum and pons.
(3) thalamus, quadrigemina hypothalamus and corpora.
(4) amygdala, hippocampus and corpus callosum.

200. Variations caused by mutation, as proposed by Hugo de Vries are:
(1) random and directional.
(2) random and directionless.
(3) small and directional.
(4) small and directionless.

■■■